

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Chemistry
2. **Subject Name:** Inorganic Chemistry: Principle, Structure and Reactivity
L-T-P: 3-1-0 Credits: 4
3. **Pre-requisites:** Organometallics
4. **Syllabus and reference books:**

Syllabus:

1. Bonding in transition-metals complexes: Introduction to symmetry and group theory
2. L-S coupling for d^n states and terms, splitting of levels and terms in chemical environment for high spin and low spin complexes (octahedral, tetrahedral, and square planer environment). Orgel and Tanabe-Sugano diagrams.
3. Ligand field theory, molecular orbital theory: Projection operator, Ligand group of orbitals, MOT of octahedral complexes with only sigma bond and sigma bond and π bond, MOT of ferrocene.
4. Electronic transitions: Laporte selection rule, spin selection rule. Vibronic coupling, calculation of $10Dq$ from electronic spectra.
5. Magnetic properties: Curie law, van Vleck equation, 1st order and 2nd order Zeeman effect, paramagnetic, ferromagnetism, antiferromagnetism.
6. Inorganic reaction mechanisms-: Substitution reaction, electron transfer reaction, photochemical reactions.

Reference Books:

- (1) Ligand Field theory and applications: B.N. Figgis and Michael A. Hitchman, 1999
- (2) Inorganic Chemistry-Principles of Structure and Reactivity. J. E. Huheey, E. A. Keiter, R. L. Keiter, 4th Edn. Harper-Collins, NY, **1993**.
- (3) Inorganic Chemistry: Weller, Overton, Rourke, Armstrong. Oxford, 6th Edition
- (4) Introduction to Ligands Fields, B. N. Figgis, 1966
- (5) Chemical Applications of Group Theory, 3rd Ed, F. Albert Cotton, 2008
- (6) Inorganic Chemistry by G.L Miessler and Donald A Tarr, 5th edition, 2014.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Bonding in transition-metals complexes: Introduction to symmetry and group theory	5 h
2.	L-S coupling for d^n states and terms, splitting of levels and terms in chemical environment for high spin and low spin complexes (octahedral, tetrahedral, and square planer environment). Orgel and Tanabe-Sugano diagrams	8 h
3.	Ligand field theory, molecular orbital theory: Projection operator, Ligand	8 h

	group of orbitals, MOT of octahedral complexes with only sigma bond and sigma bond and π bond, MOT of ferrocene.	
4.	Electronic transitions: Laporte selection rule, spin selection rule. Vibronic coupling, calculation of 10Dq from electronic spectra	5 h
5.	Magnetic properties: Curie law, van Vleck equation, 1 st order and 2 nd order Zeeman effect, paramagnetic, ferromagnetism, antiferromagnetism	6 h
6.	Inorganic reaction mechanisms-: Substitution reaction, electron transfer reaction, photochemical reactions	10 h
7.	Tutorial and Assignments	10 h
Total number of hours		52 h